EEEN20054 – Torque in circular shafts

➢ This report should not exceed 6 pages; do not change the formatting settings.

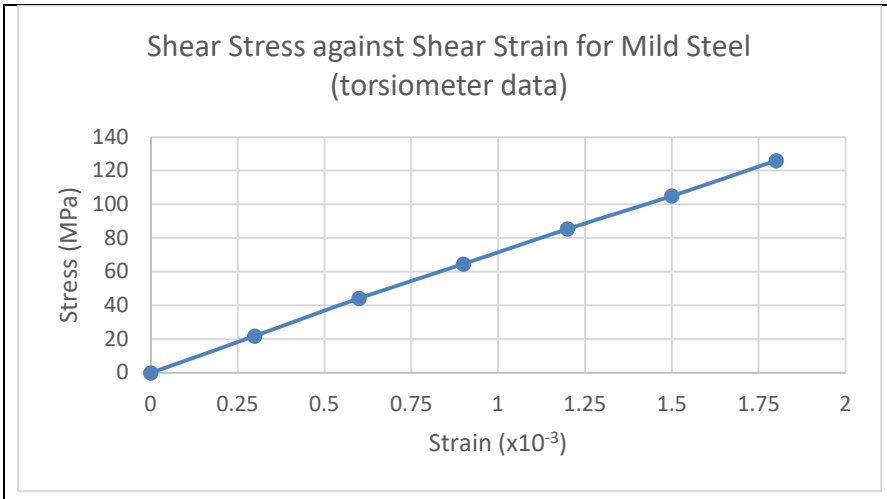
1. Measurements with torsiometer [25 marks]

1.1. Table 1; measured and calculated values

Value of I_p =	$1.27 \times 10^{-10} m^4$

Torsiometer gauge reading (inches)	Twist angle of torsiometer (deg)	Torque (Nm)	Shear Stress (MPa)	Shear Strain
0.000	0.000	0.000	0.000	0.000
0.005	0.286	0.930	21.9	2.99×10^{-4}
0.010	0.573	1.880	44.3	6.00×10^{-4}
0.015	0.859	2.740	64.6	9.00×10^{-4}
0.020	1.146	3.620	85.4	1.200×10^{-3}
0.025	1.432	4.460	105	1.500×10^{-3}
0.030	1.719	5.330	126	1.800×10^{-3}

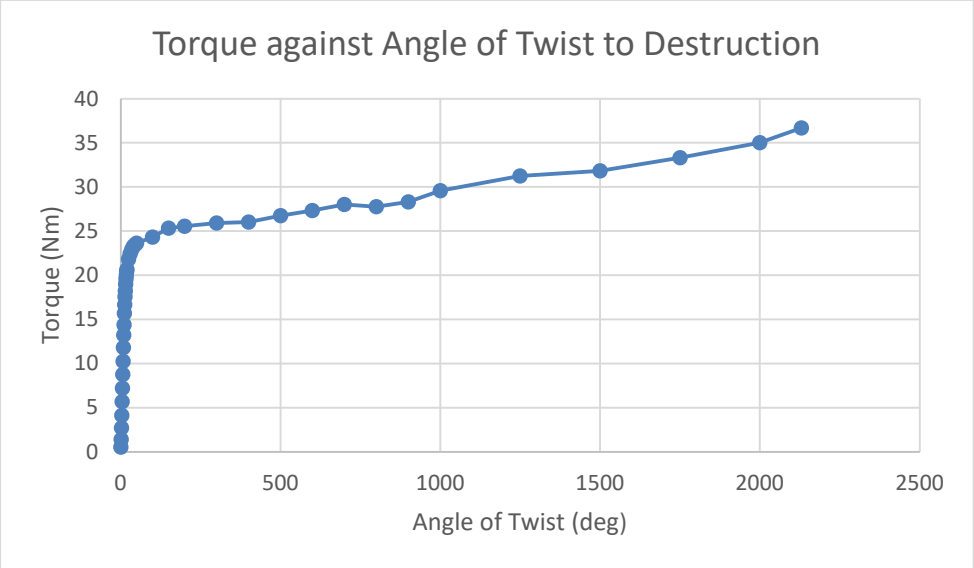
1.2. Plot of shear stress against shear strain, torsiometer

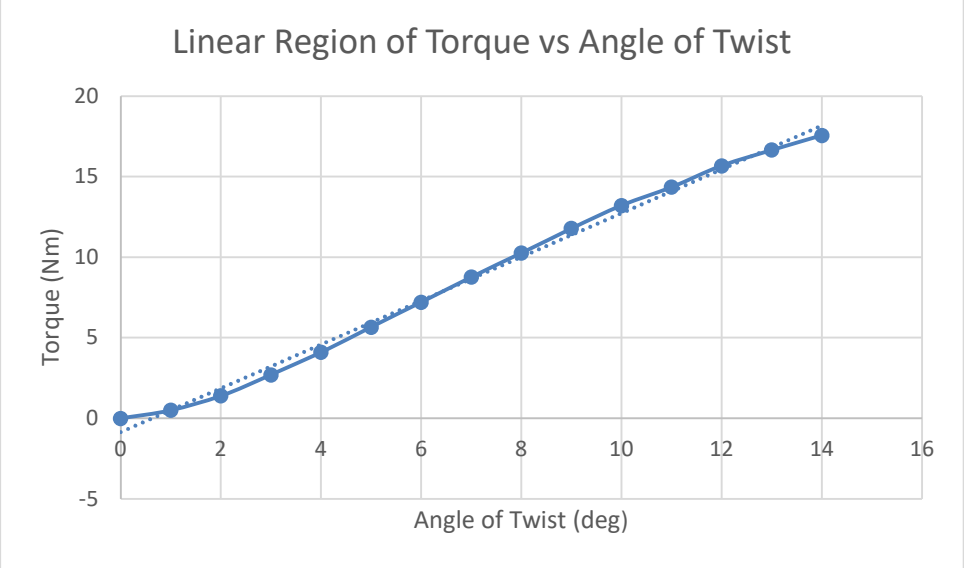


The gradient produced a shear modulus of elasticity of 69.7 GPa. This slight difference with this value and the typical expected value could be attributed to minor experimental errors such as friction in the torsiometer dial gauge

Shear modulus of elasticity, G from plot:	69.7 GPa

2. Plots of Torque against angle of twist, test to destruction [20 marks]

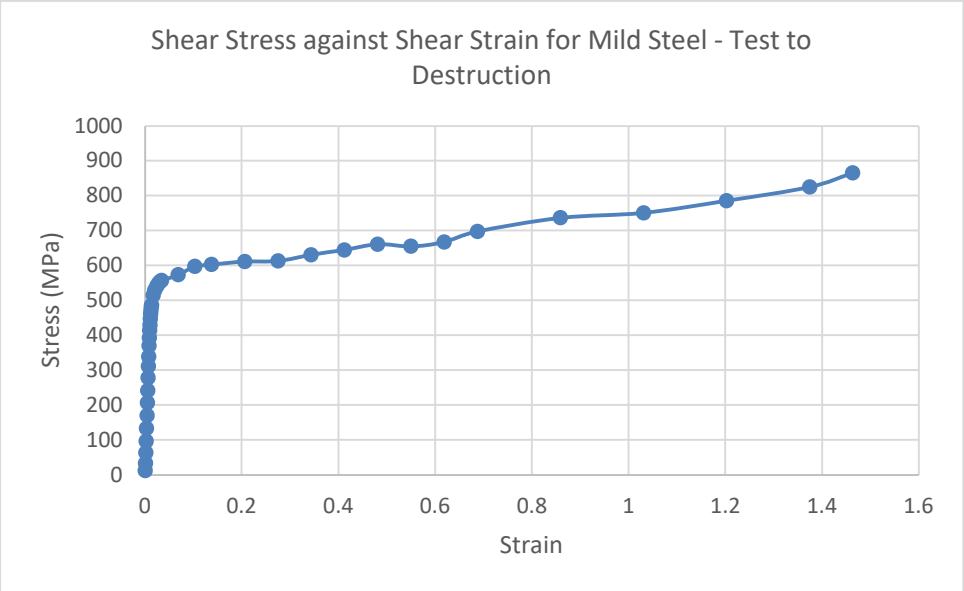


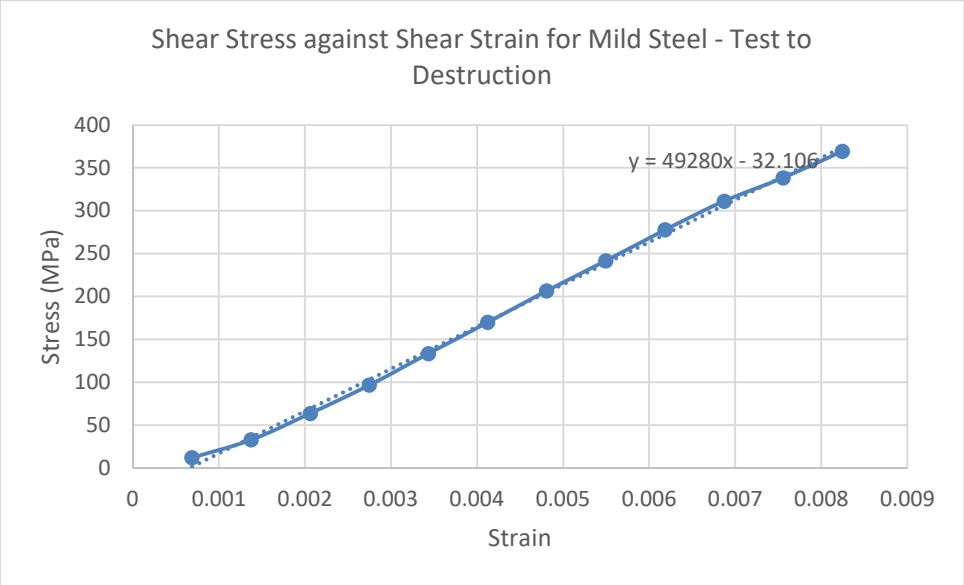


From the torque vs. angle of twist graph, the linear elastic region extends up to approximately 13° of twist. This was determined by plotting the initial portion of the torque-twist curve, fitting a linear trend through the origin and identifying the point where the measured torque deviated from the linear prediction. The corresponding torque at the proportional limit is approximately 16.66 Nm. beyond this torque, the relationship between torque and twist becomes nonlinear, showing signs of plastic deformation. Therefore, to ensure deformations remain reversible (elastic) the applied torque should not exceed 16.66 Nm.

Maximum allowable torque for the shaft such that deformations are reversible. (Justify your answer in the textbox above)	16.66 Nm

3. Plots of shear stress against shear strain, tests to destruction [30 marks]





Similar to part 2, the shear modulus was determined from the linear elastic region of the shear stress-shear strain graph. The linear trendline fitted to the data points up to approximately 0.0085 strain corresponding to 49280 MPa

- Using the 0.2% offset method and the last linear strain value of 0.00824565, the yield stress can be calculated as follows: = 49280 x (0.00824565 – 0.002) = 308 MPa
- However, the last clear linear part on the graph shows a yield stress of approximately 370 MPa. The 0.2% offset method value was used for this report.

Shear modulus of elasticity, G from plot:	49.3GPa
Shear yield stress of the shaft (analogous to the tensile yield stress)	308 MPa

4. **Other results** [25 marks]

Comment on the difference between the two measured values of G obtained for the shaft:

The torsiometer had a G value of 69.7 GPa, whereas the destruction test had a G value of 49.3 GPa for mild steel. This difference could be attributed to the sensitivity and accuracy of the measuring equipment. The torsiometer is a specialised instrument designed to measure specifically for small, precise measurements within the elastic range by measuring twist directly on a 50mm gauge length specimen while minimising the influence of external factors (slack in the drive system). In contrast, the destruction test measures the rotation of the entire specimen (76.2mm), including the gearbox and drive sockets. This method also involves much larger strains of up to 0.0085, which introduces more friction. These factors can lead to a lower measured stiffness and, therefore a lower calculated value of G.

Young Modulus of Shaft, E = 2 x G x (1 + v)

With G = 49.3 GPa (destruction test): E = 2 x 49.3 x 1.3 = 128 GPa

With G = 69.7 GPa (destruction test): E = 2 x 69.7 x 1.3 = 181 GPa

To conclude, the torsiometer value is closer to the expected Young's modulus for mild steel of approximately 200-210 GPa, showing it is more reliable and accurate as it avoids friction effects at larger strains and uses shorter, more controlled gauge length

Young's modulus of the shaft (assume Poisson's ratio is 0.3):	181 GPa (G = 69.7)